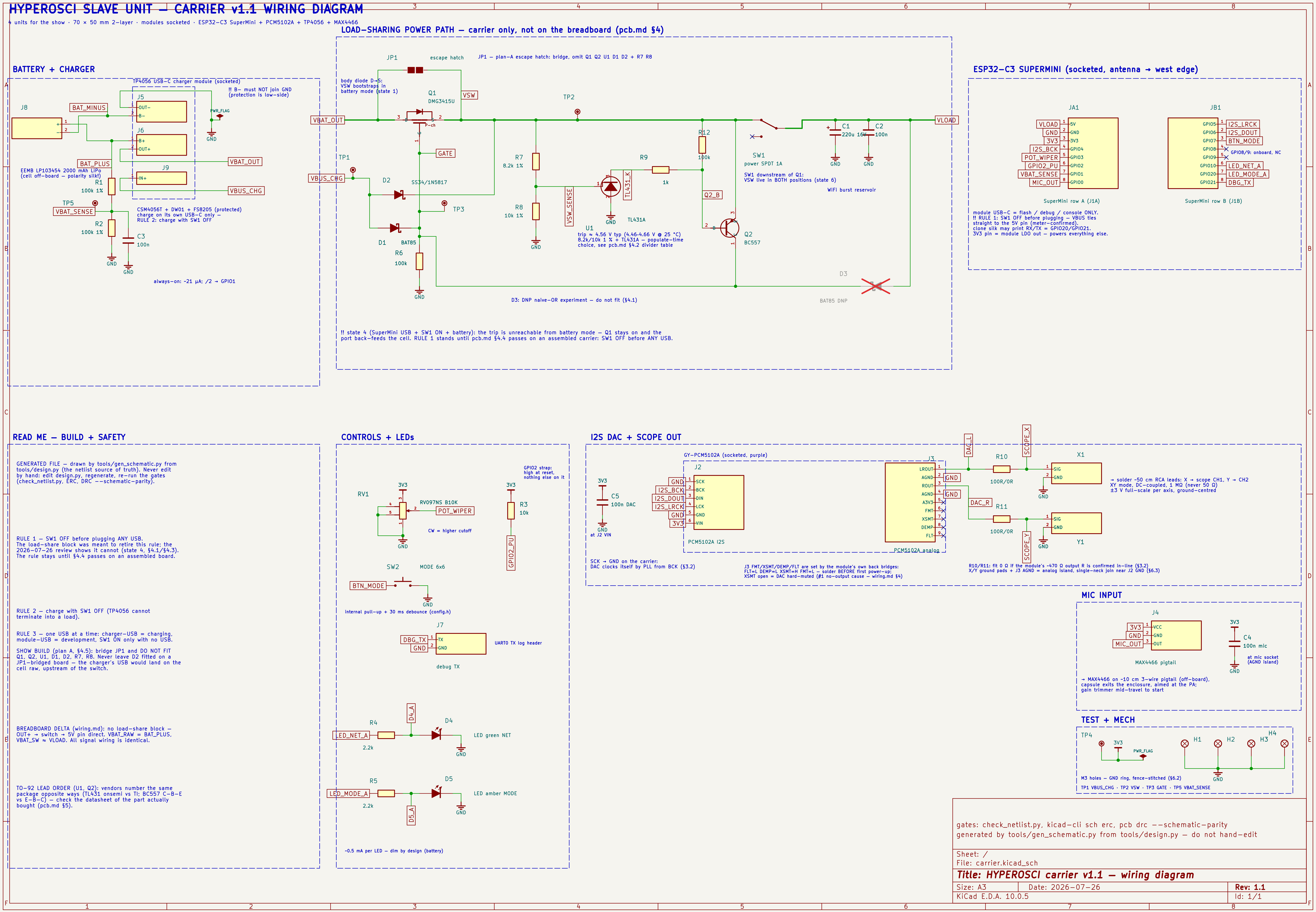
HYPEROSCI SLAVE UNIT — CARRIER v1.1 WIRING DIAGRAM
4 units for the show · 70 × 50 mm 2-layer · modules socketed · ESP32-C3 SuperMini + PCM5102A + TP4056 + MAX4466
BATTERY + CHARGER
TP4056 USB-C charger module (socketed)
!! B- must NOT join GND (protection is low-side)
EEMB LP103454 2000 mAh LiPo (cell off-board — polarity silk!)
CSM4056T + DW01 + FS8205 (protected) charge on its own USB-C only — RULE 2: charge with SW1 OFF
always-on: ~21 µA; /2 → GPIO1
LOAD-SHARING POWER PATH — carrier only, not on the breadboard (pcb.md §4)
JP1 — plan-A escape hatch: bridge, omit Q1 Q2 U1 D1 D2 + R7 R8
body diode D→S: VSW bootstraps in battery mode (state 1)
SW1 downstream of Q1: VSW live in BOTH positions (state 6)
WiFi burst reservoir
trip ≈ 4.56 V typ (4.46–4.66 V @ 25 °C) 8.2k/10k 1 % + TL431A — populate-time choice, see pcb.md §4.2 divider table
D3: DNP naive-OR experiment — do not fit (§4.1)
!! state 4 (SuperMini USB + SW1 ON + battery): the trip is unreachable from battery mode — Q1 stays on and the port back-feeds the cell. RULE 1 stands until pcb.md §4.4 passes on an assembled carrier: SW1 OFF before ANY USB.
ESP32-C3 SUPERMINI (socketed, antenna → west edge)
SuperMini row A (J1A)
SuperMini row B (J1B)
GPIO8/9: onboard, NC
module USB-C = flash / debug / console ONLY.
!! RULE 1: SW1 OFF before plugging — VBUS ties straight to the 5V pin (meter-confirmed).
clone silk may print RX/TX = GPIO20/GPIO21.
3V3 pin = module LDO out — powers everything else.
READ ME — BUILD + SAFETY
GENERATED FILE — drawn by tools/gen_schematic.py from tools/design.py (the netlist source of truth). Never edit by hand: edit design.py, regenerate, re-run the gates (check_netlist.py, ERC, DRC --schematic-parity).
RULE 1 — SW1 OFF before plugging ANY USB. The load-share block was meant to retire this rule; the 2026-07-26 review shows it cannot (state 4, §4.1/§4.3). The rule stays until §4.4 passes on an assembled board.
RULE 2 — charge with SW1 OFF (TP4056 cannot terminate into a load).
RULE 3 — one USB at a time: charger-USB = charging, module-USB = development, SW1 ON only with no USB.
SHOW BUILD (plan A, §4.5): bridge JP1 and DO NOT FIT Q1, Q2, U1, D1, D2, R7, R8. Never leave D2 fitted on a JP1-bridged board — the charger's USB would land on the cell raw, upstream of the switch.
BREADBOARD DELTA (wiring.md): no load-share block — OUT+ → switch → 5V pin direct. VBAT_RAW = BAT_PLUS, VBAT_SW ≈ VLOAD. All signal wiring is identical.
TO-92 LEAD ORDER (U1, Q2): vendors number the same package opposite ways (TL431 onsemi vs TI; BC557 C-B-E vs E-B-C) — check the datasheet of the part actually bought (pcb.md §5).
CONTROLS + LEDs
GPIO2 strap: high at reset, nothing else on it
CW = higher cutoff
internal pull-up + 30 ms debounce (config.h)
UART0 TX log header
debug TX
LED green NET
LED amber MODE
~0.5 mA per LED — dim by design (battery)
I2S DAC + SCOPE OUT
GY-PCM5102A (socketed, purple)
SCK → GND on the carrier: DAC clocks itself by PLL from BCK (§3.2)
J3 FMT/XSMT/DEMP/FLT are set by the module's own back bridges: FLT=L DEMP=L XSMT=H FMT=L — solder BEFORE first power-up; XSMT open = DAC hard-muted (#1 no-output cause — wiring.md §4)
→ solder ~50 cm RCA leads: X → scope CH1, Y → CH2 XY mode, DC-coupled, 1 MΩ (never 50 Ω) ±3 V full-scale per axis, ground-centred
R10/R11: fit 0 Ω if the module's ~470 Ω output R is confirmed in-line (§3.2) X/Y ground pads + J3 AGND = analog island, single-neck join near J2 GND (§6.3)
MIC INPUT
MAX4466 pigtail
at mic socket (AGND island)
→ MAX4466 on ~10 cm 3-wire pigtail (off-board), capsule exits the enclosure, aimed at the PA; gain trimmer mid-travel to start
TEST + MECH
M3 holes — GND ring, fence-stitched (§6.2)
TP1 VBUS_CHG · TP2 VSW · TP3 GATE · TP5 VBAT_SENSE
gates: check_netlist.py, kicad-cli sch erc, pcb drc --schematic-parity
generated by tools/gen_schematic.py from tools/design.py — do not hand-edit
Sheet: /
File: carrier.kicad_sch
Title: HYPEROSCI carrier v1.1 — wiring diagram
Size: A3
Date: 2026-07-26
Rev: 1.1
KiCad E.D.A. 10.0.5
Id: 1/1